

PORTFLOW

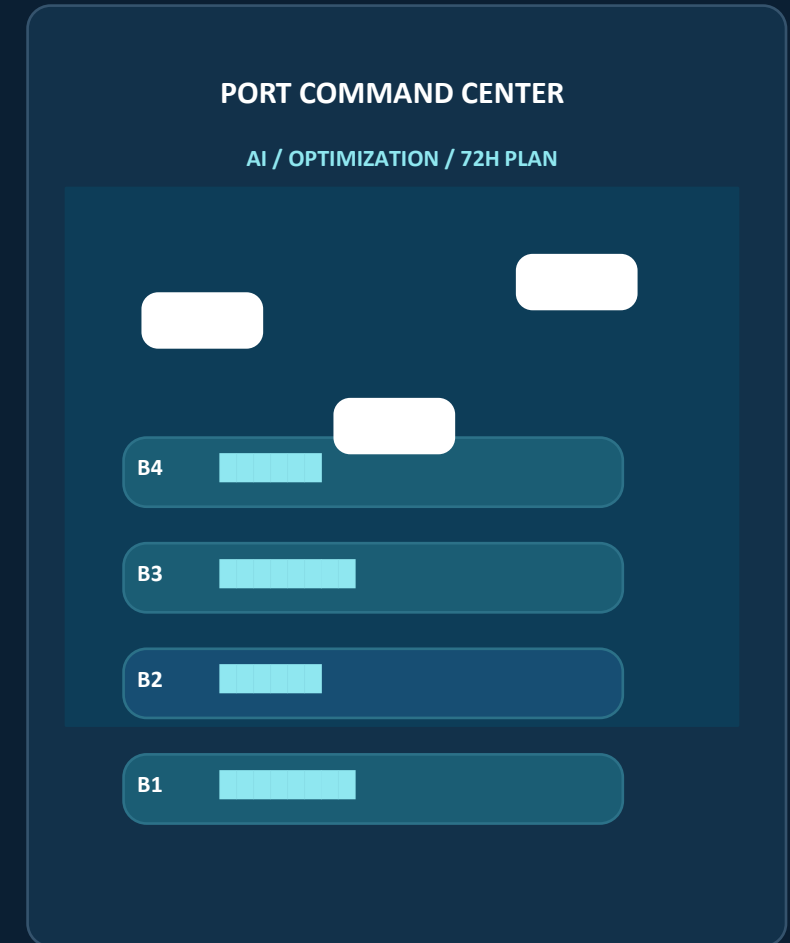
Container Congestion Predictor & Port Operations Optimiser

Predict. Optimize. Act.

CRITICAL NOW • SMART PORT & LOGISTICS OPERATIONS

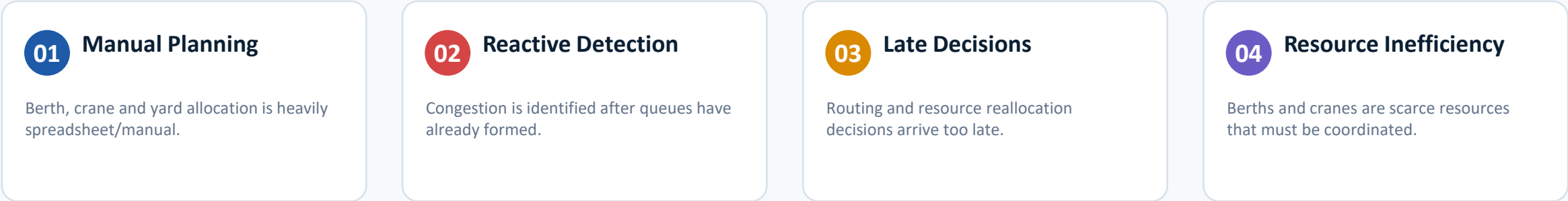
Team: HackHarvest

Built as an MVP decision-support system for proactive port operations.

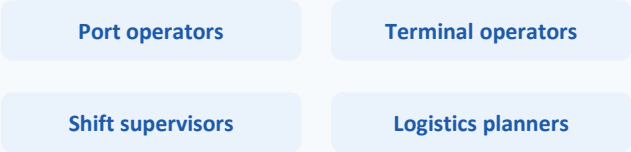


Problem — Port Congestion Is Detected Too Late

Who has it, why it matters, and what breaks in today's workflow



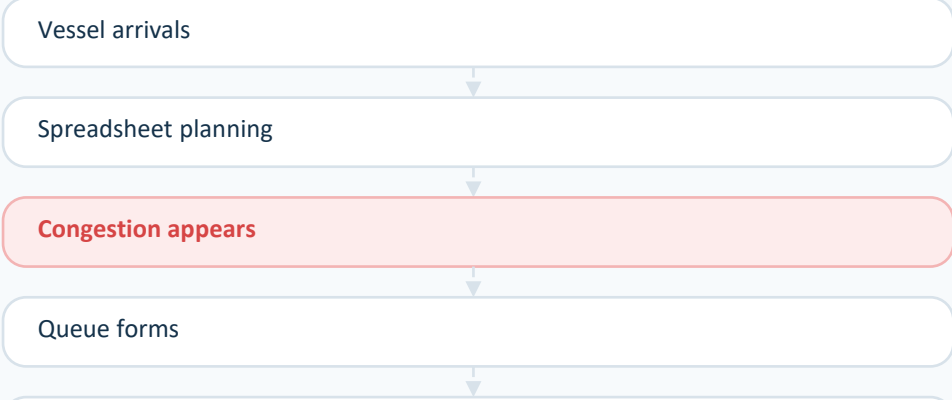
WHO HAS THE PROBLEM?



WHY IT MATTERS



CURRENT WORKFLOW



Ports need to move from reactive planning to predictive, constraint-aware decision-making.

Solution — PortFlow: Predict → Optimize → Act

One system turns port data into an actionable 72-hour operating plan

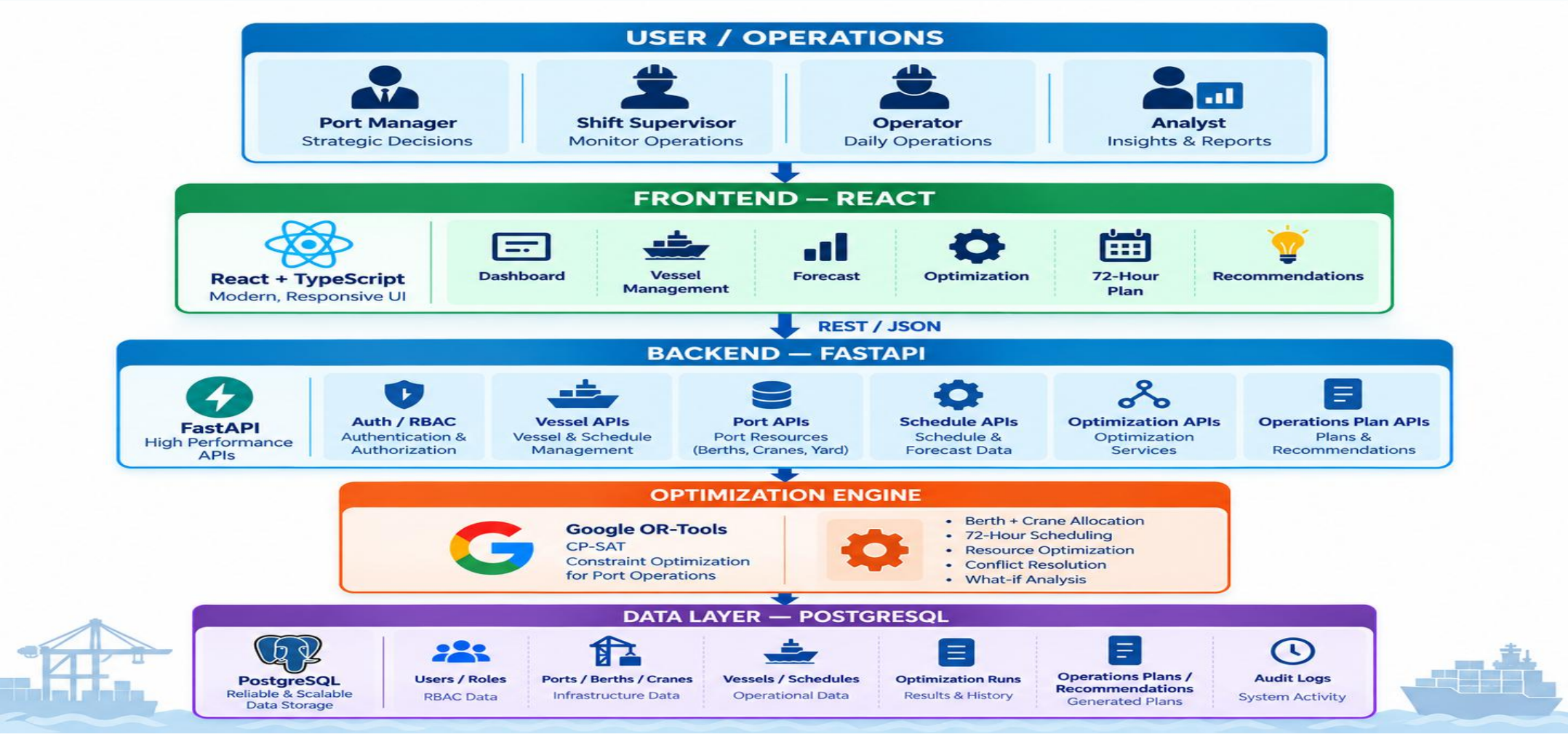


TRADITIONAL Spreadsheet → Reactive → Queue → Late Decision

iSOLUTION Predict → Optimize → Recommend → Plan

Architecture — How the System Works Technically

React + FastAPI + AI/Optimization services + PostgreSQL



Architecture deliberately keeps prediction and optimization as separate services so the MVP can evolve without rewriting the API layer.

Demo — 72-Hour Port Operations Planner

Best feature: predict congestion, optimize resources, and generate a supervisor-ready plan

ACTIVE VESSELS

24

BERTH UTILIZATION

82%

CRANE UTILIZATION

74%

CONGESTION RISK

HIGH

HORIZON

72 HRS

72-HOUR OPERATIONS PLAN

15 SEP

16 SEP

17 SEP

B1

V101

B2

V102

B3

V103

B4

V104

B5

V105

Shift 1

Shift 2

Shift 3

CRITICAL ALERT

Berth B2 projected to reach high utilization.

Recommendation

Move V104
B2 → B4

Impact

Expected waiting time ↓

DEMO FLOW Load schedules → Predict risk → Run optimizer → Review recommendation → Publish 72-hour plan

IBM Technologies — How They Fit

Keep this slide transparent: show implemented IBM components only when actually integrated

IBM TECHNOLOGY

ROLE IN ISOLUTION

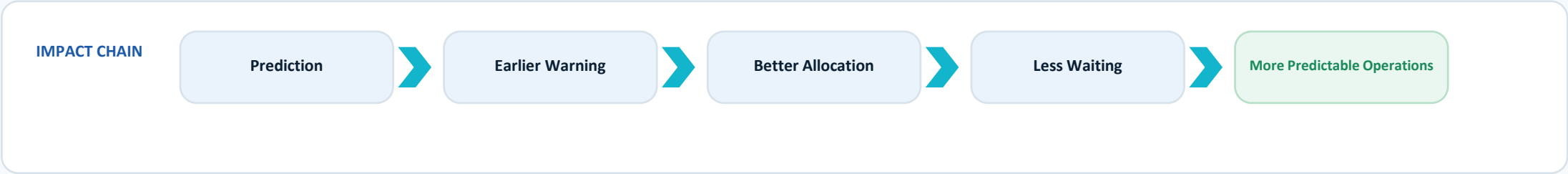
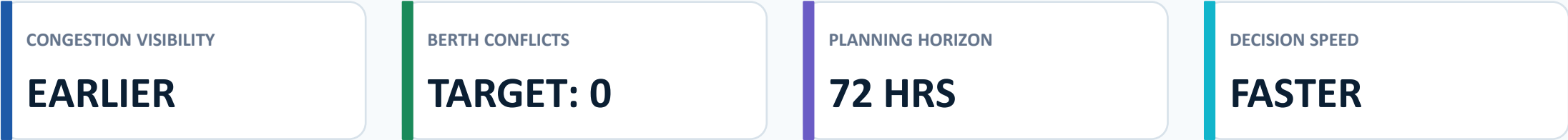
STATUS

IBM watsonx / watsonx.ai	AI model experimentation, evaluation, and future prediction workflows	PLANNED / TARGET
IBM Cloud	Cloud deployment and infrastructure option for production	PLANNED / TARGET
IBM managed data services	Managed database / data-platform option if selected	PLANNED / TARGET
Core MVP stack	React + FastAPI + PostgreSQL + OR-Tools	CURRENT MVP

IBM integration principle Use IBM services where they add measurable value — AI, cloud, model lifecycle, or managed data.

Results — What Success Looks Like

For MVP, focus on operational outcomes and measurable demo evidence



- 01 Operational** Identify high-risk congestion before queue formation.
- 02 Optimization** Produce feasible berth + crane assignments under constraints.
- 03 Supervisor** Turn system output into an actionable 72-hour operating plan.
- 04 Scalability** Extend later with AIS, weather, routing, and deeper ML.

Team — Building the Port of Tomorrow

Assign clear ownership across product, engineering, AI, optimization, and demo delivery

1

Patel Ankit

Backend / API Engineer

FastAPI • DB • Auth • RBAC

2

Patel Yug

Backend/API Engineer

Congestion prediction • features

3

Solanki Aryant

Optimization Engineer

OR-Tools • berth/crane constraints

4

Dhruv Seth

Frontend / Integration Engineer

React • dashboard • 72h plan UI

